Question:
Carlos is planting a lemon tree. The tree will cost $90 to plant. Each year it will grow 7 lemons, which he can sell for $1.5 each. It costs $3 a year to water and feed the tree. How many years will it take before he starts earning money on the lemon tree?

Reasoning + Answer:

<th inking process >
Step 1 : We know the initial cost of planting the tree is $ 9 0 . Carlos does not earn anything in the first year because he has to pay for planting and the tree hasn 't started producing le mons yet .

Step 2 : Each year , the tree produces 7 le mons . Each lemon sells for $ 1 . 5 , so the annual revenue from le mons is :
7 le mons ÃĹ $ 1 . 5 = $ 1 0 . 5 0 per year .

Step 3 : Each year , it costs $ 3 to water and feed the tree .

So , the net profit (or loss) per year after planting costs is :
Annual revenue âĢĵ Annual cost = $ 1 0 . 5 0 âĢĵ $ 3 = $ 7 . 5 0 per year .

But we need to determine when Carlos starts earning money âĢĶ that is , when the cumulative earnings from the le mons (after subtract ing water and feed costs) exceed the initial planting cost of $ 9 0 .

Step 4 : We need to find the number of years , n , such that the total net profit (after subtract ing the initial $ 9 0 cost) becomes positive .

Total net profit after n years = (n ÃĹ $ 7 . 5 0) âĢĵ $ 9 0 .

We want to find the smallest integer n such that :
n ÃĹ $ 7 . 5 0 > $ 9 0

S olve for n :
n > 9 0 / 7 . 5
n > 1 2

So , after 1 2 years , total net profit is exactly :
1 2 ÃĹ 7 . 5 0 = $ 9 0 âĨĴ breaks even .

Therefore , after 1 3 years , the profit becomes positive .

But the question is : " How many years will it take before he starts earning money on the lemon tree ?"

" Start s earning money " means the first year when he begins to make a profit âĢĶ so we need to find when the cumulative earnings exceed the initial cost .

So after 1 2 years , he has just broken even (net profit = 0). He does not earn money yet .

After 1 3 years , he has earned $ 7 . 5 0 more than the initial cost âĨĴ he starts earning money in year 1 3 .

Therefore , it takes 1 3 years to start earning money .

But wait âĢĶ does " start earning money " mean the first year when he makes a profit , or when the cumulative profit becomes positive ?

Yes âĢĶ " start earning money " means the first year when he makes a profit . Since in year 1 3 , he has made a profit of $ 7 . 5 0 (after covering the initial cost), and he did not earn money in years 1 through 1 2 , the answer is 1 3 years .

Final check :
Year 1 to 1 2 : total net profit = 1 2 ÃĹ 7 . 5 0 = $ 9 0 âĨĴ breaks even âĨĴ no profit .
Year 1 3 : profit = $ 7 . 5 0 âĨĴ positive profit âĨĴ starts earning money .

So , after 1 3 years , he starts earning money .

< answer > 1 3 </ answer >